



MMSeg: A novel multi-task learning framework for class imbalance and label scarcity in medical image segmentation

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ABSTRACT

Semi-supervised medical image segmentation can effectively alleviate the high costs associated with obtaining high-quality labels and issues related to data privacy. However, the issue of imbalance in medical images limits the practical application of semi-supervised frameworks. One of the core principles of semi-supervised learning is to infer and predict by learning data distributions. Since unlabeled samples lack explicit class information, both inter-class and intra-class imbalances in the regions of interest can undermine the training core of this framework. To alleviate these issues, we propose a segmentation framework based on multi-task learning, which integrates both self-supervised and semi-supervised learning components within the multi-task framework. In the self-supervised component, we utilize four sub-tasks to extract the internal structure and features of images from unlabeled data, aimed at mitigating the imbalance issues in regions of interest. In the semi-supervised learning component, we employ a segmentation network with a teacher–student model structure that effectively generates high-confidence pseudo-labels from unlabeled data, thereby enhancing the model's ability to recognize minority classes. To maintain the robustness of the model, we introduce a progressive learning strategy and a dynamic weight adjustment mechanism to prevent feature representation bias caused by varying task difficulties. Experimental results indicate that our proposed method achieved the best performance in extensive experiments on two abdominal multi-organ segmentation tasks, outperforming other competitive methods. Codes are available at <https://github.com/ReinforceLove/MMSeg>.

1. Introduction

Medical image segmentation technology is vital for accurately identifying and extracting relevant areas from intricate medical imagery, essential for disease diagnosis, treatment planning, and condition monitoring. It is well known that acquiring medical images involves high costs and privacy issues, the quantity and quality of both the source images and their corresponding labels limit the application of fully supervised methods. Therefore, effectively using a small amount of labeled data to train a good segmentation model has become a promising direction, attracting widespread attention in the medical imaging community. To address this issue, researchers have proposed various semi-supervised methods for medical image segmentation. However, one significant challenge in practical applications is the imbalance

problem for a learning model. This includes both input imbalance and output imbalance [1], for input imbalance, small foreground elements interspersed in an abundance of background voxels, in addition to regions of interest (ROI) with varying dimensions; for output imbalance, the occurrence of false positive and false negative samples is inevitable during the inference process. Ideally, we aim to eliminate both types of erroneous samples, but in reality, as one type of error decreases, the other tends to increase. This is particularly evident in medical image segmentation tasks involving specific pathological types or rare diseases. As shown in Fig. 1, in multi-organ segmentation, the foreground ratios of different organs demonstrate a high degree of imbalance. In the case of lesion segmentation, the difference between

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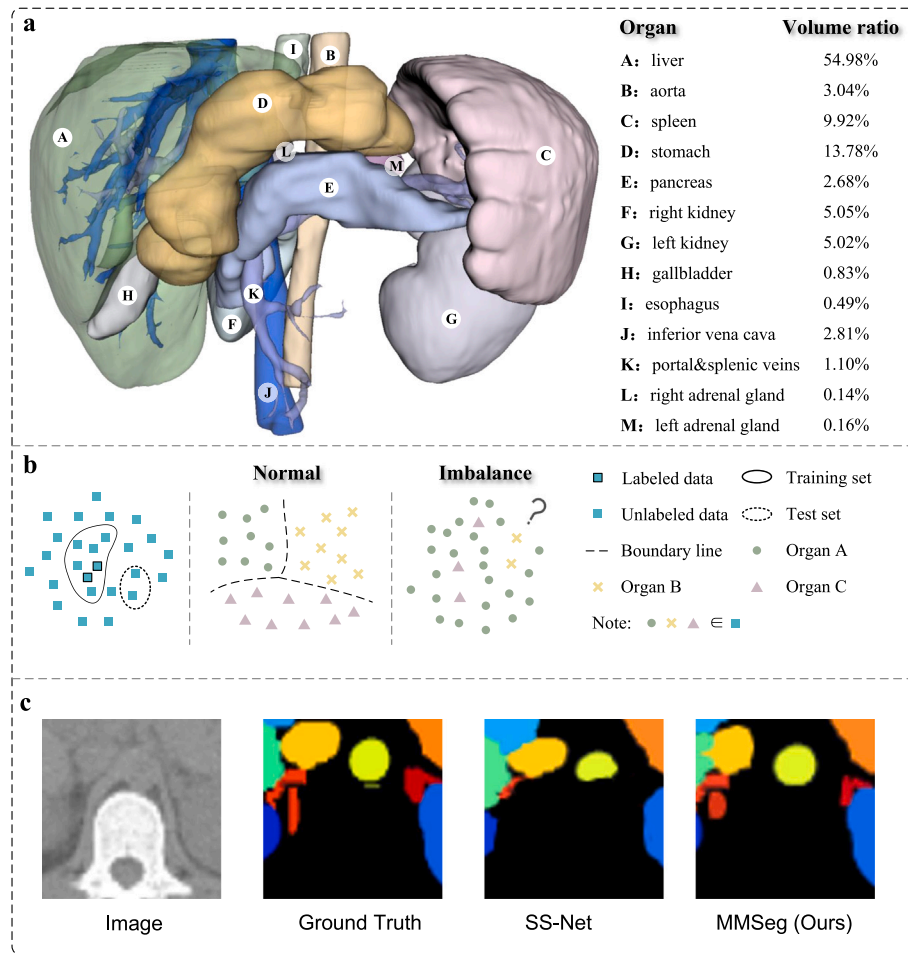


Fig. 1. Illustration of the imbalanced multi-organ segmentation. In Fig. 1a, we demonstrate the morphologies and volume differences of 13 different organs [2]. The volume differences among various classes can exceed 20-fold. Additionally, without any cropping, the volume difference between the smallest foreground element and the entire background element can exceed 1000-fold. In Fig. 1b, we show the difference in training under class-balanced and class-imbalanced conditions using semi-supervised learning. Ideally, the decision boundary should be located in areas with fewer samples. However, in imbalanced data, these areas may not be accurately identified. Due to the limited annotations, semi-supervised learning cannot directly estimate the overall class information from the labels, nor can it construct the class weights or sampling methods for the samples based on the labels. In Fig. 1c, we utilized specific examples (AMOS2022) to provide a visual and detailed illustration of the impact of data imbalance on segmentation performance.

foreground elements can exceed 20-fold, and the contrast between foreground and background may exceed 1000-fold if the images are not pre-trimmed.

Currently, fully supervised learning methods for imbalanced medical image segmentation can generally be categorized into reweighting-based approaches or resampling-based approaches. Reweighting methods address the class imbalance by adjusting the weights of each category in the loss function of the model. During the training of deep learning models, each category can be assigned a weight inversely proportional to its frequency of occurrence. This means that less frequent categories are given higher weights in the loss calculation, while more common categories receive lower weights. This strategy increases the model's focus on minority classes, thereby enhancing the segmentation accuracy for these categories. Resampling addresses class imbalance by altering the number of samples for each category in the training dataset. This typically involves oversampling the minority classes (increasing the number of minority-class foreground voxels) and undersampling the majority classes (reducing the number of majority-class background voxels), or a combination of both. Although these approaches have achieved incremental success, they struggle to adapt to scenarios with insufficient labels. Some researchers introduced semi-supervised methods to address issues of imbalance. For instance, Lin et al. [3] addressed the class imbalance in knee MRI images by reweighting the loss function and adjusting the cropping probabilities based on label distribution. Wang et al. [4] proposed

a novel semi-supervised learning framework that used distribution-aware debiased weighting and difficulty-aware debiased weighting to address data imbalance by dynamically leveraging pseudo labels. Basak et al. [5] employed dynamic confidence arrays to obtain class sampling rates, addressing biases in semi-supervised learning.

However, these methods focused on addressing imbalance issues have certain limitations when integrated with a semi-supervised framework. In reweighting methods, the main issue is how to weigh the losses for these unlabeled samples. Since unlabeled samples lack explicit class information, it is difficult to determine how to set weights for them, making it hard to accurately assess the true distribution of classes across the entire dataset. Therefore, directly applying reweighting strategies based on class frequency from fully supervised learning may not accurately reflect the actual situation of the data. In resampling methods, not all samples have explicit labels, so resampling cannot be directly based on label information, leading to challenges in effectively utilizing these unlabeled data. Ignoring these data may result in the loss of important information, while blindly using them could introduce noise. Adjusting the training set's class distribution through resampling, such as oversampling minority classes or undersampling majority classes, may alter the true distribution of the data, which is particularly significant in semi-supervised learning.

To tackle these issues, we observe the potential of multi-task learning, which can be considered a form of shared representation learning.

This shared mechanism enhances the model's ability to learn general data features, particularly advantageous when data labels are incomplete or class distributions are imbalanced, improving recognition of minority classes. Furthermore, within a semi-supervised learning framework, multi-task learning enables the utilization of labeled data for one task and the employment of unlabeled data for additional tasks.

Along these lines, we propose a multi-task medical image segmentation method, the MMSeg. Our multi-task learning framework integrates self-supervised auxiliary tasks with a semi-supervised main task. Specifically, the self-supervised tasks include rotation prediction [6], masked inpainting [7], image denoising [8], and contrastive learning [9], which encourage the model to capture the internal structure and features of input images. The semi-supervised task is designed to enhance the performance of the segmentation network on limited labeled data. During training, the self-supervised and semi-supervised tasks share encoder parameters. To ensure balance between tasks, we employ a dynamic weighting mechanism to adjust the loss weights. Additionally, we adopt a progressive learning strategy that initially focuses on using self-supervised tasks to build feature representations, gradually introducing semi-supervised tasks to optimize segmentation performance. By combining self-supervised and semi-supervised learning tasks, our model can effectively learn data representations from unlabeled data and make fine adjustments using limited labeled data, ultimately achieving superior segmentation results.

To summarize, our contributions are shown as follows:

- Our method combines self-supervised auxiliary tasks with semi-supervised main tasks to effectively increase the model's efficiency in utilizing unlabeled data. Self-supervised tasks extract useful features from this data, helping the model capture the intrinsic structure of medical images and enhancing its ability to generalize to unseen samples. This is especially important for medical image segmentation, where acquiring precisely labeled images can be challenging.
- We propose a multi-task learning framework to address the imbalance between foreground and background, as well as among different foreground elements in medical images. The deep and diverse feature representations provided by self-supervised learning offer a more balanced perspective for the semi-supervised learning tasks, thereby improving overall segmentation outcomes.
- We introduce a dynamic weighting mechanism and a progressive learning strategy to mitigate the issue of feature representation bias towards any one task due to varying task convergence speeds in multi-task learning. This approach reduces negative interference between tasks, providing an effective methodology and technical support for future applications in other medical imaging tasks.

In Section 2, we delve into the related research work on the task of imbalanced medical image segmentation. In Section 3, we provide a detailed introduction to the proposed model, including its workflow, various modules, and technical details. The experimental results and ablation studies are presented in Section 4. Finally, the conclusion and future work are outlined in Section 5.

2. Related work

In this section, we explore recent work on our method from three perspectives: (1) semi-supervised imbalance medical image segmentation methods; (2) imbalance medical image analysis; and (3) multi-task learning frameworks.

2.1. Semi-supervised imbalance medical image segmentation methods

Existing semi-supervised methods provide valuable insights into how to address label scarcity and imbalance in medical image segmentation, which inspire our approach by demonstrating effective pseudo-

label generation and class reweighting strategies. Semi-supervised learning methods enable segmentation models to exploit rich label information from unlabeled images, which is crucial in few-shot medical image segmentation where annotated data is limited. Prominent approaches in this field predominantly employ two frameworks: (1) the teacher–student framework [10], where the student model initially trains on labeled data, and subsequently a teacher model is formed through the student's Exponential Moving Average (EMA). This teacher model generates pseudo labels that are used to further enhance the student model's training in a synergistic feedback loop; (2) the Cross Pseudo Supervision (CPS) framework [11], which exploits consistency between two perturbed models, using the pseudo label generated by one to train the other, thereby optimizing learning outcomes. However, these methods face limitations in extremely imbalanced datasets and when pseudo-label quality is compromised due to poor initial predictions, which motivates us to explore more robust strategies for both pseudo-label refinement and effective learning under severe imbalance.

Researchers proposed semi-supervised methods to address the imbalance in medical images, broadly categorized into four streams: (1) To address the issues of inter-class and intra-class reweighting, researchers proposed strategies such as maintaining confidence arrays, loss weighting, and dual-debiasing heterogeneous collaborative training frameworks [4,5,12]. Prototype-based methods offer a complementary approach by leveraging representative prototypes to enhance feature discrimination, particularly for small-target lesions, where samples are extremely limited [13]. These methods improve segmentation performance by prioritizing poorly performing classes and optimizing loss functions. However, they may still face limitations when dealing with extremely imbalanced datasets, such as when some rare classes have very few samples (small-target lesion segmentation). Even with weighting and prioritization, it remains difficult to obtain sufficient samples for effective learning. (2) For improving pseudo-label quality, advancements have been made by integrating corrected pseudo-supervision and voxel-level contrastive learning, as well as self-correcting cooperative and contrast-enhanced co-training models [14,15]. Generative models such as GMMSeg [16] further enhance pseudo-label quality by modeling the feature distribution with Gaussian mixtures, enabling more robust and reliable pseudo-label generation for unlabeled data. These approaches significantly enhance the learning objectives for unlabeled data. However, the generation of pseudo-labels depends on the initial prediction quality of the unlabeled data. Poor initial pseudo-label quality can lead to error propagation and impact final segmentation performance. While self-correcting methods alleviate this issue to some extent, generating high-quality initial pseudo-labels remains a challenge. (3) To address issues of low contrast and blurred edges, some methods emphasize pixel-level smoothness and inter-class separability, along with bidirectional copy-pasting between labeled and unlabeled data to reduce the empirical distribution gap [17,18]. Yet, these methods may still perform poorly in complex real-world scenarios, especially in highly noisy and intricate backgrounds. (4) The use of large models (such as SAM) to improve pseudo-label quality has also shown promising results, further enhancing the performance of semi-supervised medical image segmentation [19,20]. However, large models typically require substantial computational resources and storage, limiting their widespread application in practical settings.

2.2. Imbalance medical image analysis

Imbalance medical image analysis methods tackle the challenges of uneven data distributions, providing foundational approaches that align with our goal of improving small-target lesion segmentation under extreme imbalance scenarios. The characteristic of imbalanced medical images lies in the significant disparities in inter-class or intra-class representations of regions of interest. For organ segmentation, the natural differences in the size and shape of different organ tissues make it challenging to segment multiple organs within a fixed-sized window.

For lesion segmentation, normal tissue typically predominates, while pathological tissues are less prevalent. This imbalance between foreground and background makes it difficult to segment small target areas of interest, even when there is only one type of lesion present [21].

To mitigate class imbalance by specialized loss functions, Nasalwai et al. [22] designed the Accelerated Tversky loss to handle class imbalances. Similarly, Zhao et al. [23] proposed Focal Dice loss to address the imbalance between hard and easy examples, leading to better segmentation outcomes. In the case of framework innovations, Yang et al. [24] employed contrastive learning to enhance feature representation, addressing imbalanced data distributions. Wang et al. [4] proposed the Dual-debiased Heterogeneous Co-training (DHC) framework to leverage pseudo labels in semi-supervised 3D medical image segmentation to improve performance significantly. To reduce the dependency on large annotated datasets, Ouyang et al. [25] introduced a self-supervised few-shot semantic segmentation framework. This method used superpixel-based pseudo-labels, effectively addressing the imbalance problem and eliminating the need for annotations. For multi-scale network architectures, Xia et al. [26] introduced the 2.5D multi-path Transformer fusion network (MTr-Net) to integrate Z-axis information to address boundary deformation and contextual residual imbalance, thereby improving segmentation accuracy for organs and lesions. Nonetheless, existing methods often struggle with capturing fine-grained features of rare classes or adapting to highly imbalanced datasets, underscoring the need for more adaptive and robust solutions to these challenges.

2.3. Multi-task learning methods

Multi-task learning provides a flexible framework that informs our method by demonstrating how shared feature representations can mitigate issues of label scarcity and class imbalance in medical image segmentation. Multi-task Learning is a machine learning approach that enhances the generalization ability of models by learning across multiple related tasks simultaneously. This method is predicated on the core idea that different tasks can share useful information, thereby fostering mutual learning benefits. In a multi-task learning framework, a shared neural network base is typically designed to learn common features across all tasks, while the top layers are specialized for specific tasks. This structure improves performance on individual tasks, increases adaptability to new tasks, and reduces the risk of overfitting [27].

In medical image processing, we discovered that multi-task Learning has the potential to address issues of class imbalance and label scarcity. Multi-task Learning enables the model to share valuable feature representations by simultaneously training on several related tasks. This process enhances the model's ability to extract a more complete set of information from the data [28]. For instance, Huang et al. [29] improved organ segmentation accuracy in medical images by leveraging anatomical prior information and a multi-task learning approach to fully utilize unlabeled data.

Multi-task learning emphasizes enhancing the expressiveness of features by learning the correlations between tasks [30]. In different medical image segmentation tasks, such as organ segmentation and lesion detection, may focus on different objects, but the underlying image features (such as texture, shape, and edges) might share commonalities. Through multi-task learning, the model can optimize the expression of these common features while learning specific tasks, thereby improving the model's ability to interpret complex medical images.

In imbalanced medical image segmentation, one task might focus on identifying common lesions, while another task focuses on less common types. This shared learning framework allows the model to leverage features associated with the majority class when dealing with minority classes, thereby improving the accuracy and generalization ability to identify minority classes. Regarding the issue of label scarcity, multi-task learning can enhance learning outcomes by utilizing unlabeled data or data with few labels [31]. For example, by integrating semi-supervised learning techniques, one task can use labeled data while

another task or multiple tasks utilize unlabeled data to learn broader feature representations. This strategy can effectively expand the scale and diversity of training samples, thereby improving the model's performance and robustness. However, challenges remain in effectively balancing task-specific objectives and ensuring the robustness of shared features, particularly in highly imbalanced and data-scarce settings, driving us to further optimize these frameworks for medical image segmentation.

3. Method

In this section, we provide a detailed overview of the proposed method outlined in Fig. 2, which includes four components: (1) a self-supervised module for helping the model capture the intrinsic structure for imbalance medical images and enhancing its ability to generalize to unseen samples; (2) a semi-supervised module to exploit rich label information from unlabeled images; (3) a progressive learning strategy for prevent potential inefficiencies and fitting difficulties during joint training; (4) dynamic weight adjustment mechanism to alleviate bias in feature representations caused by inconsistent training speeds across tasks. Then, we discuss the key components of the method and explain the network architecture along with the implementation details involved in the training and testing processes.

3.1. Overview

Our study aims to achieve two main objectives: (1) to learn representations from both labeled and unlabeled image data, to mitigate the impacts of a scarcity of high-quality labeled data; (2) to address the issue of class imbalance between foreground and background as well as between different foreground classes in medical image segmentation, thereby enhancing the effectiveness of medical image segmentation.

As shown in Fig. 2, we propose a multi-task medical image segmentation framework, named MMSeg. Our framework comprises two key components: a self-supervised learning module to uncover the robust feature representation foundation and a semi-supervised segmentation module to tackle the issue of insufficient labeled data. We assume that the dataset contains N_L labeled samples $\{(x_i^l, y_i) \mid i = 1^{N_L}\}$ and NU unlabeled samples $\{(x_i^u)_{i=1}^{N_U}\}$, where $x_i \in \mathbb{R}^{H \times W}$ represents the input image, and $y_i \in \mathbb{R}^{K \times H \times W}$ is the ground truth annotation with K classes (including background), where H and W denote the height and width of the two-dimensional images.

In the self-supervised learning module, we aim to encode the ROI information of each image data x_i to extract potential features from the data. Inspired by previous multi-task self-supervised learning [32] approaches, we designed four self-supervised tasks, including image rotation, mask inpainting, image denoising, and contrastive learning, to achieve medical image representation learning. To better achieve knowledge sharing between multiple tasks and improve the segmentation performance for minority classes, we share the encoding layer of the segmentation network in both the self-supervised tasks and the semi-supervised learning module. Simultaneously, the encoder is connected to the projection heads of the four self-supervised tasks and the segmentation network decoder, which are used for self-supervised learning and medical image segmentation, respectively. This design not only improves training efficiency but also enables the model to learn deeper and more useful feature representations from unlabeled data, including textures, edges, shapes, and other anatomical structures, thereby better supporting the performance of the segmentation network. However, we found that due to the varying difficulty levels of the self-supervised tasks, the model tends to forget previously learned knowledge when learning new tasks, leading to the problem of "Catastrophic Forgetting". To address this, we propose a gradient-based dynamic weighting mechanism. This mechanism determines the difficulty of each task by analyzing gradient updates in each iteration and accordingly adjusts the weight proportions of different tasks in the

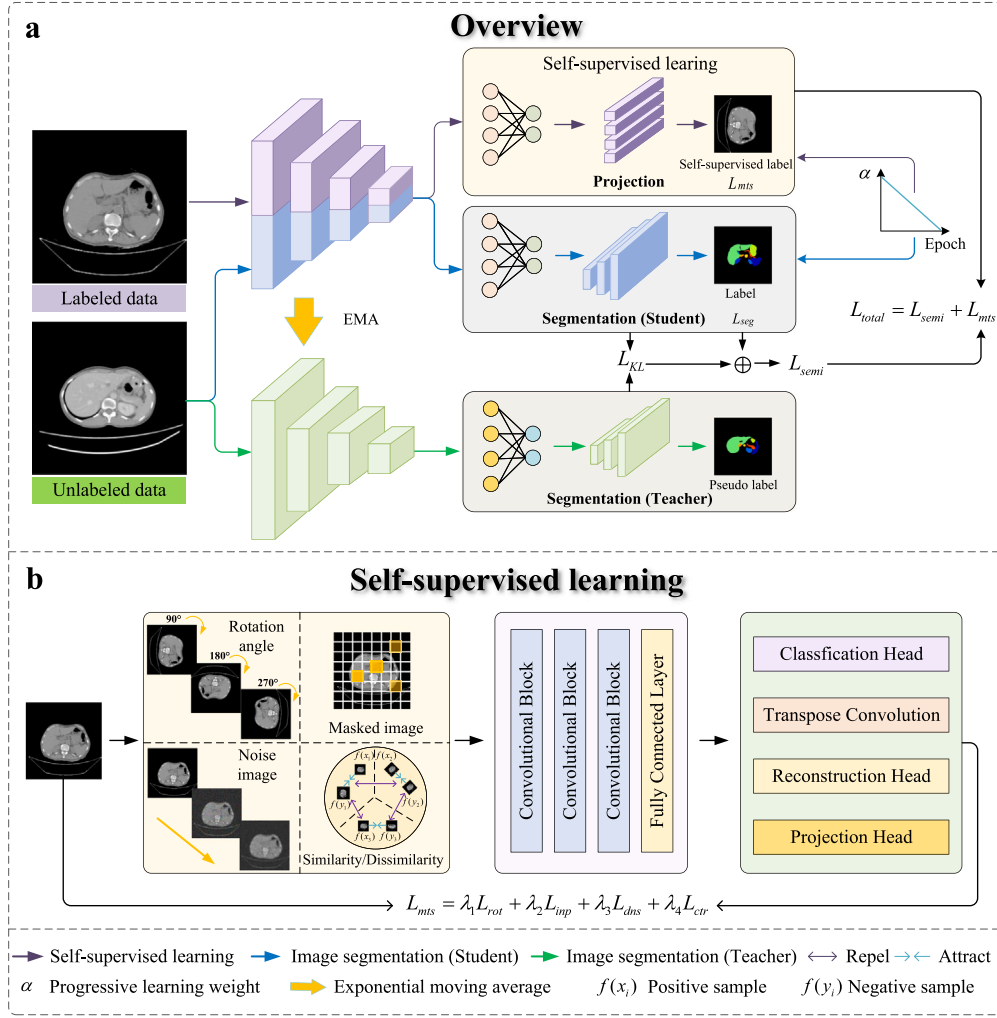


Fig. 2. (a) The overview of our proposed MMseg. Our framework employs the same encoder-decoder network structure for both the teacher and student models in the medical image segmentation task. The self-supervised learning module shares encoder weights with the student segmentation network in semi-supervised learning. Moreover, we adopt a progressive learning strategy and a dynamic weight adjustment mechanism to coordinate the convergence of multiple tasks. (b) Details the overall pipeline of the multi-task self-supervised learning module, combining four self-supervised learning tasks: rotation prediction, masked inpainting, image denoising, and contrastive learning.

overall loss function, thereby alleviating the “Catastrophic Forgetting” phenomenon.

The semi-supervised learning module leverages the shared encoder with the self-supervised learning module to effectively utilize representation information learned from the images themselves. This approach enables the model to address class imbalance issues and reduces biases towards high-frequency categories in the images. Specifically, when processing unlabeled data, the semi-supervised learning module can identify and learn critical segmentation task features while maintaining sensitivity and recognition abilities for minority classes. Additionally, we constructed a teacher-student model with identical medical image segmentation network structures. The student segmentation network shares the encoding layers with the multitask self-supervised module, using the learned representations from the self-supervised tasks to encode and segment the labeled dataset $\{(x_i^l, y_i) \mid i = 1^{N_L}\}$. The teacher model initially shares the parameters of the student segmentation network trained on the labeled dataset and then segments the unlabeled dataset $\{(x_i^u)\}_{i=1}^{N_U}$ to generate pseudo-labels. The pseudo-labels and the real labeled dataset are used as inputs for further training of the student segmentation network and the self-supervised module. During training, to achieve better feature optimization and enhance the learning of minority classes, we propose a progressive learning approach. At the beginning of the training, self-supervised learning takes the lead, and gradually, the focus shifts to segmentation network learning.

This method ensures thorough learning of feature representations while progressively enhancing the performance of the segmentation network in actual segmentation tasks.

3.2. Self-supervised learning module

In medical image segmentation tasks, the class imbalance problem among different background regions can significantly affect segmentation performance. To address this issue, we designed a self-supervised learning module to assist the semi-supervised learning module, focusing on learning the boundaries and structural features of regions of interest (ROIs) in medical images. Inspired by previous research on multi-task self-supervised learning [32,33], this module simultaneously executes multiple self-supervised tasks and the segmentation task through a shared encoder.

In the MMseg self-supervised module, we introduce a rotation prediction task that requires the model to correctly predict the rotation angle of an image. This encourages the model to understand the overall layout and main components of the image, thereby learning global structural and directional features. The occlusion restoration task enables the model to learn the local context and global consistency of the image by understanding surrounding pixel information and image structures. The image denoising task forces the model to focus on detailed features and textures within the image. The contrastive

learning task enhances the model's ability to distinguish fine details by pulling positive sample pairs closer and pushing negative sample pairs apart, thus learning effective feature representations. During training, we apply data augmentation techniques to each 2D medical image x_i to generate labels y_i^r for self-supervised learning. Finally, to ensure optimal performance of the segmentation task, the total loss L_{mts} of the self-supervised learning module is as follows:

$$L_{mts} = \lambda_1 L_{rot} + \lambda_2 L_{inp} + \lambda_3 L_{dns} + \lambda_4 L_{ctr}, \quad (1)$$

where λ_i represents the weights for different self-supervised tasks, and L_{rot} , L_{inp} , L_{dns} , and L_{ctr} denote the loss functions for rotation prediction, masked inpainting, image denoising, and contrastive learning, respectively. The following sections provide a detailed explanation of these four tasks.

3.2.1. Image rotation

We introduce a rotation prediction task as part of our multi-task self-supervised learning, specifically designed to predict the rotational angle categories of medical images accurately. This task enables the learning of the orientation and structural features inherent in the images. Initially, the images are rotated around their center points to four fixed angles: 0° , 90° , 180° , and 270° . These rotated angles serve as labels for the self-supervised learning task, denoted as y_i^r . The decoder for rotation prediction consists of a multilayer perceptron classifier, which outputs the softmax probabilities for each rotation category, represented as $\hat{y}_i^r$. To evaluate model performance and optimize it, we use the cross-entropy loss function L_{rot} to measure the effectiveness of the rotation prediction task:

$$L_{rot} = - \sum_{r=1}^R y_i^r \log(\hat{y}_i^r), \quad (2)$$

where R represents the total number of rotational classes being predicted. Rotating images is also employed as a data augmentation method in contrastive learning.

3.2.2. Masked inpainting

To enhance the identification of key structures and features within the region of interest (ROI) in medical images, our MMseg model integrates a mask repair task into its self-supervised learning module. Specifically, we segment the image into 16×16 patches and randomly mask 30% of the patches within the ROI. To reconstruct these masked areas, we introduce a transposed convolutional layer as a reconstruction head to rebuild the entire image. The reconstructed image output is denoted as $\hat{x}_i^{mask}$. The goal of this process is to minimize the $L1$ loss between the original image x_i and the reconstructed image $\hat{x}_i^{mask}$.

$$L_{inp} = \|x_i - \hat{x}_i^{mask}\|_1, \quad (3)$$

where L_{inp} represents the loss function for the masked inpainting task. This metric quantifies the absolute differences pixel-by-pixel, aiming to effectively restore the masked portions and accurately recover the underlying details in the ROI.

3.2.3. Image denoising

The primary goal of the image denoising task is to eliminate noise interference from images to restore a clearer image that more closely resembles its original state. We can effectively enhance the model's ability to recognize details and edge information in medical image analysis, thereby improving the model's capability to distinguish between foreground and background categories. In our method, MMseg employs data augmentation techniques, specifically by adding Gaussian white noise with a standard deviation of 0.01 to the original medical images x_i , simulating the types of noise that might be encountered in actual medical imaging processes. The images with added noise are labeled as $x_i^{L_{dns}}$. In the image denoising decoding stage, we have selected a transposed convolution module, which excels in restoring image

details. Additionally, we use the Root Mean Square Error (RMSE) loss function L_{dns} to optimize and assess the image denoising performance:

$$L_{dns} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i^{L_{dns}})^2}, \quad (4)$$

where $\hat{x}_i^{L_{dns}}$ are the pixel values of the denoised image, n is the total number of pixels in the image.

3.2.4. Contrastive learning

In previous research [34,35], we observed that contrastive learning exhibited excellent performance in addressing class imbalance issues. We employed data augmentation techniques to expand the sample set to further enhance the model's generalization capabilities. The core mechanism of contrastive learning involves maximizing the mutual information between positive pairs while minimizing the mutual information between negative pairs, thereby enabling the model to capture subtle differences between images. This approach effectively increases the similarity within the same category of samples and amplifies the differences between different categories. It allows the model to recognize and differentiate between categories more accurately, effectively addressing the challenge of class imbalance. In our shared encoder E_{srl} , each augmented image is mapped to a latent representation v , and we use cosine similarity as the distance measure for encoded representations. Formally, the contrastive loss between V_i and V_j is defined as:

$$L_{ctr} = - \log \frac{\exp(\text{sim}(v_i, v_j) / \tau)}{\sum_k^{2N} 1_{k \neq i} \exp(\text{sim}(v_i, v_k) / \tau)}, \quad (5)$$

where τ is the measurement of the normalized temperature scale. 1 is the indicator function that evaluates to 1 if and only if $k \neq i$. sim denotes the dot product between normalized embeddings. The MMseg model enhances intra-class compactness and inter-class separability through L_{ctr} .

3.3. Semi-supervised learning module

The semi-supervised learning module aims to learn the feature distribution of medical images through limited labels and unlabeled data. This module consists of a segmentation network for medical image segmentation and a teacher-student model for generating pseudo-labels. The segmentation network learns the latent representations of the images by sharing encoder parameters with the self-supervised learning module. The teacher-student model assigns higher confidence weights to minority classes when generating pseudo-labels, forcing the model to pay more attention to these minority classes. During training, the initial parameters of the teacher model are obtained by training on the labeled data using the segmentation network and the self-supervised module.

3.3.1. Medical image segmentation

Our segmentation task shares the same encoder E_{srl} , with the self-supervised learning module. This design allows the model to transfer and utilize shared feature representations across different tasks. Through this approach, the encoder can capture image features that are beneficial for all tasks, thereby improving the model's performance on each task and enhancing the recognition and segmentation of minority classes.

Specifically, when the encoder receives a 2D image x_i , where $x \in \mathbb{R}^{C \times H \times W}$ and C is the number of feature channels, it reduces the spatial resolution of the feature maps through convolution and downsampling processes, while simultaneously increasing the depth of the feature maps to capture higher-level semantic information. After each convolution and downsampling, the spatial dimensions of the feature maps are halved, resulting in $x \in \mathbb{R}^{C' \times \frac{H}{2} \times \frac{W}{2}}$, where C' is the number of feature channels post-convolution. In the segmentation task, the

decoder uses three layers of upsampling to progressively generate the predicted segmentation result $\hat{y} \in \mathbb{R}^{C \times H \times W}$. Additionally, by employing skip connections, the decoder can integrate high-resolution features from the encoder during the upsampling process, which aids in learning across different semantic levels. The Dice loss function L_{seg} is used to evaluate segmentation performance:

$$L_{seg} = 1 - \frac{2 \times I + \varepsilon}{U + \varepsilon}, \quad (6)$$

where I is the intersection between the predicted result $\hat{y}_i$ and the true label y_i , representing the total number of pixels classified as positive in both. U is the union of the predicted result $\hat{y}_i$ and the true label y_i , representing the total number of pixels classified as positive in at least one of the labels. ε is the smoothing coefficient.

3.3.2. Teacher–student model structure

In the semi-supervised learning module, we found that due to the limited amount of labeled data, there is insufficient supervision signal for minority class samples. Therefore, we introduced a teacher–student model architecture based on the medical image segmentation network. By utilizing unlabeled data to generate pseudo-labels and assigning higher confidence weights to minority classes, we force the model to focus more on minority classes during the learning process. Specifically, in the MMSeg framework, we first count the minority class weights W_{mc} in the labeled dataset $\{(x_i^l, y_i^l)\}_{i=1}^{N_L}$, then train the model on this data and use it as the teacher model. Next, the unlabeled dataset $\{x_i^u\}_{i=1}^{N_U}$ is input into the teacher model to generate pseudo-labels y_i^u . The quality of these pseudo-labels is evaluated based on confidence scores σ_i , and higher weights are assigned to the minority class labels to ensure that the model focuses more on the minority classes during the training process. Lastly, both labeled and unlabeled data are input into the MMSeg framework for training. During the training process, the student model updates the teacher model's parameters through the Exponential Moving Average (EMA) mechanism, ensuring a smooth parameter update process. This reduces fluctuations in the teacher model's parameters, making it more stable and reliable when generating pseudo-labels. The formula is as follows:

$$\theta_T^{(k+1)} = \alpha \theta_T^{(k)} + (1 - \alpha) \theta_S^{(k)}, \quad (7)$$

where, $\theta_T^{(k+1)}$ represents the parameters of the teacher model after the $k + 1$ iteration, $\theta_T^{(k)}$ represents the parameters of the teacher model after the k iteration, and $\theta_S^{(k)}$ represents the parameters of the student model after the k iteration. The coefficient α is the EMA factor, which ranges from $[0, 1]$. Finally, through multiple self-training iterations, the teacher–student model utilizes the L_{total}^{semi} loss for learning and adjustment, thereby optimizing the segmentation performance of minority classes, as follows:

$$D_{KL}(q||p) = \sum_c q(c|x) \log \left(\frac{q(c|x)}{p(c|x)} \right), \quad (8)$$

$$L_{KL} = \sum_{x \in \text{Pixels}} D_{KL}(q(x)||p(x)), \quad (9)$$

$$L_{semi} = \begin{cases} L_{seg} & \text{if } k = 0 \\ L_{seg} + L_{KL} & \text{if } 0 < k < K, \end{cases} \quad (10)$$

where $D_{KL}(q||p)$ represents the KL divergence for category c at a given pixel point x , measuring the difference between the probability distributions output by the teacher model q and the student model p . $q(c|x)$ is the probability predicted by the teacher model that a given pixel x belongs to category c . $p(c|x)$ is the probability predicted by the student model that a given pixel x belongs to category c . L_{KL} is the total KL divergence loss for the image, which is the sum of the KL divergence losses for all pixel points. k is the current training iteration number, and K is the total number of iterations.

3.4. Progressive learning strategy

In the MMSeg framework, the self-supervised learning module and the semi-supervised learning module work in synergy. The self-supervised learning module extracts latent representations of medical image data through multi-task self-supervised learning, exploring the intrinsic distribution features of the data. The semi-supervised learning module leverages the features extracted by the self-supervised learning module, combined with labeled and unlabeled data, to generate pseudo-labels with weighting for minority classes. However, previous research [36] indicates that joint training multi-task self-supervised learning and segmentation tasks may lead to auxiliary tasks dominating the training process, making it difficult to balance the features of each task effectively, thus reducing the model's generalization ability.

To address this challenge, we devised a progressive learning strategy. Initially, the self-supervised module takes the lead in training, enabling more effective learning of the intrinsic representation information embedded in the images themselves. As training progresses, the segmentation network gradually begins to leverage the representation information learned through the self-supervised module to optimize the segmentation of medical images. We introduced a parameter β to dynamically adjust the learning dynamics between the multi-task module and the semi-supervised learning module, gradually allowing the semi-supervised learning task to assume a leading role as follows:

$$\beta = \frac{\text{iterations}}{\text{Max_iterations}} \quad (11)$$

$$L_{total} = (1 - \beta) \cdot L_{mfs} + \beta \cdot L_{semi}, \quad (12)$$

where iterations represents the current iteration count, and Max_iterations represents the total number of iterations.

3.5. Dynamic weight adjustment mechanism

We encounter inconsistent convergence speeds due to varying task difficulties in the self-supervised learning module. This inconsistency can result in some tasks being fully trained while others are still in progress. As a consequence, when the model adopts new tasks, it might overwrite previously learned knowledge, leading to a phenomenon known as “catastrophic forgetting”. To address this, we have designed a Multi-task Dynamic Loss Weight Adjustment Mechanism. This mechanism evaluates the training status of different tasks by analyzing their gradient updates and dynamically adjusts the multi-task loss weights λ_i to optimize the overall learning process. Our goal is to reduce interference between tasks, enhance the ability of the model to learn new tasks and retain the knowledge of previously trained tasks through this method.

For the dynamic loss weight adjustment mechanism, we have N different tasks with varying loss function weight coefficients λ_i . To standardize these weights, we ensure that the sum of all weight coefficients equals 1, as follows:

$$\lambda_{total} = \sum_{i=1}^N \lambda_i = 1, \quad (13)$$

The dynamic weight adjustment mechanism allows us to maintain a balance between tasks during the training process, ensuring that no single task exerts undue influence on the overall training. Additionally, we employ a gradient-based method to assess the magnitude of the gradient of the loss function for each task. The essence of this approach is that if the gradient of a task's loss updates rapidly, it indicates that the task has made significant learning progress. In such cases, to maintain balance in multi-task learning, we should appropriately reduce the weight of that task to prevent it from exerting excessive influence on the overall learning process. Firstly, we compute the L_2

norm of the gradient of the loss function L_{task} for the model parameters θ as shown:

$$\|\nabla_{\theta} L_{\text{task}}\|_2 = \sqrt{\sum_i \left(\frac{\partial L_{\text{task}}}{\partial \theta_i} \right)^2} \quad (14)$$

where the gradient norm $\|\nabla_{\theta} L_{\text{task}}\|_2$ represents the L2 norm of the gradient of the loss function L_{task} with respect to the shared network parameters θ . It provides a quantitative measure of the speed or intensity of L_{task} change in the parameter space. To calculate this norm, one must first compute the partial derivatives $\frac{\partial L_{\text{task}}}{\partial \theta_i}$ of L_{task} for each parameter θ_i , then sum the squares of these derivatives, and finally take the square root. Next, we calculate the average gradient norm G_{avg} across all tasks, which reflects the overall sensitivity of the model to all tasks:

$$G_{\text{avg}} = \frac{1}{N} \sum_{i=1}^N \|\nabla_{\theta} L_{\text{task}}\|_2, \quad (15)$$

where G_{avg} denotes the average of the L2 norms $\|\nabla_{\theta} L_{\text{task}}\|_2$ across all tasks, serving as a threshold for the current gradient updates. This average gradient norm is utilized to adjust the weights of the current tasks, ensuring a balanced and dynamic learning process that adapts to the varying rates of progress among different tasks. Finally, we dynamically adjust the loss weight λ_i for each task based on the ratio of its gradient norm to the average gradient norm.

$$\lambda_i = \left| \frac{\|\nabla_{\theta} L_{\text{task}}\|_2}{G_{\text{avg}}} - 1 \right| * \lambda_i, \quad (16)$$

We achieved the balance between tasks during the training process through a dynamic weight adjustment mechanism, ensuring that the model effectively learns all tasks while avoiding excessive influence from any single task. This strategy helps improve the model's generalization ability and learning efficiency.

3.6. Implementation details

Our approach is implemented using the PyTorch framework, and all experiments were conducted on four NVIDIA GeForce RTX 4090 GPUs with 24 GB of memory each, as well as two Intel processors. In the multi-task self-supervised module of the MMSeg framework, we construct self-supervised labels using data augmentation techniques such as rotation, masking, and adding 0.1 noise. During training, we use the Adam optimizer with an initial learning rate of 0.0008, which linearly decays as training progresses. In the semi-supervised learning module, we use a U-net-based encoder-decoder as the initial segmentation model. Additionally, we apply data augmentation to the labeled data. During training, the input image size is 256×256 , and we use the Adam optimizer with an initial learning rate of 0.0002 to iteratively train the segmentation network. Subsequently, we perform iterative training of the teacher-student model in a self-training manner.

4. Experimental results

In this section, We conducted extensive experiments on the Synapse [2] and AMOS [37] datasets for class-imbalanced semi-supervised medical image segmentation. We qualitatively and quantitatively analyzed the advantages of our proposed method compared to other methods and evaluated the effectiveness of each key component through ablation studies.

4.1. Datasets

We conducted benchmark experiments on class-imbalanced semi-supervised medical image segmentation using the Synapse [2] and AMOS [37] datasets. The Synapse dataset includes 13 foreground classes: spleen (Sp), right kidney (RK), left kidney (LK), gallbladder (Ga), esophagus (Es), liver (Li), stomach (St), aorta (Ao), inferior vena

cava (IVC), portal & splenic veins (PSV), pancreas (Pa), right adrenal gland (RAG), and left adrenal gland (LAG), along with one background class, comprising 30 axial contrast-enhanced abdominal CT scans. Compared to Synapse, the AMOS dataset eliminates PSV but adds three new categories: duodenum (Du), bladder (Bl), and prostate/uterus (P/U). We divided the aforementioned two datasets into training, validation, and testing sets in an 8:1:1 ratio.

During the data preprocessing process, we first cropped the original CT scan images to ensure that only the regions of interest (ROI) were preserved, and we reoriented all scan images to the RAS direction. Next, we normalized the CT scan images using a $[-300, 300]$ intensity window, adjusting their value range to $[0, 1]$. Then, through range scaling techniques, we adjusted the value range of all scans to $[-1, 1]$. Finally, we sliced the abdominal volumetric images along the z-axis and resampled them to a resolution of 256×256 for training the network.

4.2. Evaluation metrics

In the experimental evaluation of MMSeg, we use the Dice Similarity Coefficient (DSC) as the primary performance metric. By measuring the overlap between the predicted and actual regions, the DSC provides a quantitative assessment of the segmentation results; a high DSC value not only indicates a larger area of overlap and excellent segmentation performance but also generally signifies lower false positive and false negative rates.

4.3. Comparison with other state-of-the-art methods

To evaluate the effectiveness of the MMSeg framework in scenarios where labeled data is scarce and unlabeled data are abundant for class-imbalanced image segmentation tasks, we conducted comparative experiments using the Synapse Datasets and AMOS Datasets. In the Synapse Datasets, we randomly selected 20% or 30% of the samples, and in the AMOS Datasets, we selected 5% or 10% of the samples, ensuring that these samples included all organ categories, thereby establishing the labeled datasets. The unlabeled datasets consisted of the remaining data. For thorough evaluation, we benchmarked the MMSeg framework against the fully supervised medical image segmentation method U-net [38] and all state-of-the-art (SOTA) semi-supervised medical segmentation learning techniques, including SSNet [17], DHC [4], ICL [39], ACTION++ [40], and MiDSS [41]. All methods were implemented within the same backbone architecture framework to ensure fairness in the comparisons.

4.3.1. Comparison on the Synapse datasets

In experiments conducted with only 20% labeled data, as shown in Table 1, the MMSeg framework demonstrated significant performance advantages in the task of class-imbalanced medical image segmentation compared to other state-of-the-art semi-supervised models. The MMSeg framework achieved an average DSC of 50.14, significantly surpassing other semi-supervised image segmentation models, thereby confirming its substantial improvements in segmentation accuracy for class-imbalanced issues. MMSeg scored the highest in most categories, especially excelling in categories with high segmentation difficulty such as the Es, Li, and Ao. Compared to the fully supervised model U-Net, which had an average DSC lower by 13.24, this indicates that MMSeg still possesses significant advantages with a limited amount of labeled data. This is attributed to the model's self-supervised module for representation learning on unlabeled data and the application of the semi-supervised module's pseudo-label generation strategy.

As shown in Fig. 3, in contrast to generic semi-supervised methods like SS-Net, which perform poorly in capturing effective features of minority classes leading to subpar overall performance, MMSeg shows marked improvements in handling smaller or minority classes (Es, PSV, LAG). This also validates the effectiveness of optimizing minority class segmentation through multi-task self-supervised learning. Compared to

Table 1

Quantitative comparison between MMSeg and SOTA semi-supervised learning segmentation methods on the 20% and 30% labeled Synapse dataset. “Fully” represents fully supervised segmentation models, and some of these methods have been adapted from 3D medical image segmentation methods to 2D medical image segmentation methods. Results are reported in the ‘mean $\pm$ std’ format over 3 repeated experiments. The best results are boldfaced.

Method	Average DSC of each class (20% Labeled data)													Avg.DSC
	Sp	RK	LK	Ga	Es	Li	St	Ao	IVC	PSV	PA	RAG	LAG	
U-Net(Fully) [38]	87.2	74.0	72.5	75.2	38.8	95.4	70.3	73.9	74.7	57.2	49.6	26.9	28.3	63.38 $\pm$ 1.3
SS-Net [17]	61.3	68.4	61.2	38.3	0.0	89.6	23.1	63.2	43.2	0.0	9.3	6.2	0.0	35.68 $\pm$ 2.4
DHC [4]	61.8	69.9	60.7	65.7	15.6	86.3	34.2	67.8	63.1	32.8	31.7	32.5	10.4	48.65 $\pm$ 2.6
ICL [39]	61.5	68.6	59.8	57.4	13.9	88.2	26.3	69.1	51.2	27.6	29.3	26.7	11.3	45.45 $\pm$ 2.1
ACTION++ [40]	62.5	69.4	60.4	62.6	12.7	89.8	25.9	68.3	58.3	28.6	24.6	26.5	9.4	46.08 $\pm$ 1.6
MidSS [41]	63.3	70.8	61.1	66.2	13.9	90.1	31.5	65.3	57.4	27.4	26.5	30.7	10.5	47.28 $\pm$ 2.3
MMSeg(ours)	63.7	71.3	61.3	66.8	18.2	88.7	36.1	70.8	65.7	36.6	32.1	28.3	12.2	50.14 $\pm$ 1.8
Method	Average DSC of each class (30% Labeled data)													Avg.DSC
	Sp	RK	LK	Ga	Es	Li	St	Ao	IVC	PSV	PA	RAG	LAG	
SS-Net [17]	65.2	69.6	63.7	35.6	0.0	90.1	26.9	63.6	38.7	0.0	10.1	0.0	5.6	36.08 $\pm$ 1.9
DHC [4]	63.2	71.3	65.5	68.2	16.1	87.1	35.8	68.2	64.3	47.6	34.6	34.1	23.2	52.24 $\pm$ 0.6
ICL [39]	62.1	68.4	61.7	59.2	14.2	88.6	21.5	70.1	51.4	33.4	26.3	28.1	14.7	46.13 $\pm$ 2.0
ACTION++ [40]	63.2	70.9	62.1	64.5	13.1	90.2	27.1	68.7	58.6	28.9	25.7	21.5	8.6	46.39 $\pm$ 2.2
MidSS [41]	65.0	69.6	64.7	70.5	15.5	90.1	32.4	67.5	58.7	25.5	28.4	31.1	12.3	48.56 $\pm$ 1.7
MMSeg(ours)	66.8	72.5	64.2	68.1	18.9	89.7	38.9	70.4	57.7	38.7	35.2	29.7	31.5	52.48 $\pm$ 1.6

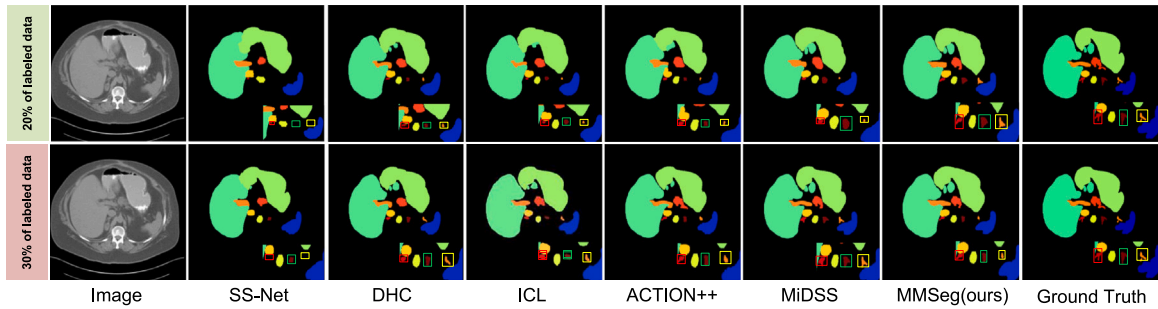


Fig. 3. Visual comparisons of segmentation results on the Synapse Datasets. The bottom-right corner of each segmentation result provides a magnified view of the segmentation details, where the red, green, and yellow boxes highlight the boundary details of the right adrenal gland (RAG), aorta (Ao), and left adrenal gland (LAG), respectively.

models like DHC and ICL that specifically address class imbalance, MMSeg’s DSC score for the RAG is 4.2 points lower than that of the DHC method. However, its overall performance remains superior. This is mainly because DHC and ICL focus on segmenting minority classes, often at the expense of the overall segmentation quality. In contrast, MMSeg utilizes a teacher–student strategy to generate pseudo-labels from unlabeled data, effectively increasing the weight of minority class labels.

When the proportion of labeled data increased to 30%, as shown in Table 1, MMSeg continued to demonstrate significant performance advantages in medical image segmentation tasks. Specifically, MMSeg achieved an average DSC score of 52.48, surpassing other comparative methods. MMSeg scored higher in several key categories, such as the Sp and RK, and was particularly outstanding in the LAG, where it scored significantly higher than other methods. Experimental results demonstrate that the MMSeg method exhibits significant advantages in dealing with class imbalances in medical image segmentation. It not only performs well with limited annotated data but also maintains a high level of segmentation effectiveness when additional annotated data is introduced, showcasing its robustness and effectiveness. By achieving high scores across various categories, MMSeg proves its adaptability and efficiency in addressing the common issue of class imbalance in medical imaging.

To further validate the performance of the MMSeg model, we conducted a visualization analysis on the Synapse dataset. As shown in Fig. 3, compared to traditional semi-supervised models, MMSeg demonstrates superior performance in segmenting minority classes and boundary areas. When the proportion of labeled data increases to 30%, the proportion of minority class samples further decreases due to the

increase in multi-class samples, causing a decline in the segmentation performance of traditional semi-supervised models for minority classes. In comparison with models such as DHC and ICL, which are specifically designed to address class imbalance, MMSeg achieves better segmentation results for minority classes. This advantage is attributed to its deeper exploration of unlabeled data, bringing the results closer to the quality of labeled data.

4.3.2. Comparison on the AMOS datasets

As shown in Table 2, experiments conducted on the AMOS dataset demonstrate that the MMSeg method achieved leading performance across various categories with only 5% of the labeled data, achieving a DSC of 50.83 ± 1.3 . This result highlights the efficiency of MMSeg in handling class imbalance in medical image segmentation tasks, particularly in key categories such as the Sp, RK, Ga, Li, St, Ao, IVC, LAG, Bl, and P/U. Additionally, MMSeg showed exceptional performance in difficult-to-segment categories such as the Es and P/U, benefiting from its unique multitask self-supervised learning mechanism that effectively extracted features from these categories while optimizing their segmentation accuracy.

When the labeled data increased to 10%, MMSeg’s overall performance improved to a DSC of 67.34 ± 1.9 , maintaining its leading position among all semi-supervised learning methods. This performance boost indicates that MMSeg can effectively utilize more labeled information as the amount of data increases, further optimizing the model’s learning and generalization capabilities. At this data annotation ratio, particularly in structurally complex categories such as the Es, Bl, and P/U, MMSeg not only maintained excellent segmentation results but also achieved significant improvements in typically challenging

Table 2

Quantitative comparison between MMSeg and SOTA semi-supervised learning segmentation methods on the 5% and 10% labeled AMOS dataset. “Fully” represents fully supervised segmentation models and some of these methods have been adapted from 3D medical image segmentation methods to 2D medical image segmentation methods. Results are reported in the ‘mean $\pm$ std’ format over 3 repeated experiments. The best results are boldfaced.

Method	Average dice of each class (5% Labeled data)															Avg.DSC
	Sp	RK	LK	Ga	Es	Li	St	Ao	IVC	PA	RAG	LAG	Du	Bl	P/U	
U-Net(Full)	93.1	92.7	94.3	68.8	69.4	94.2	81.5	92.3	84.4	73.6	58.3	61.5	62.4	66.7	59.2	76.83 $\pm$ 2.8
SS-Net	63.2	68.1	71.3	38.2	0.0	76.8	31.8	69.0	57.3	38.2	0.0	0.0	0.2	1.2	0.7	34.40 $\pm$ 1.3
DHC	67.1	70.3	70.7	42.8	38.6	76.5	44.8	71.2	58.8	46.0	17.5	27.6	26.9	40.3	34.9	48.93 $\pm$ 1.5
ICL	68.2	69.3	71.5	37.6	15.5	77.2	35.6	67.2	54.3	39.5	19.2	22.3	21.3	34.8	36.8	44.69 $\pm$ 2.3
ACTION++	67.4	68.1	69.2	35.1	21.4	78.5	42.2	68.4	61.2	27.4	14.6	16.6	23.1	37.6	31.8	44.17 $\pm$ 1.1
MidSS	66.5	69.5	70.0	36.7	28.7	79.3	43.1	73.1	59.3	37.6	13.2	14.7	24.2	35.1	32.5	45.57 $\pm$ 0.7
MMSeg(ours)	68.9	71.7	69.8	43.1	36.5	81.2	46.7	74.9	62.2	44.3	21.3	28.3	25.3	51.4	36.9	50.83 $\pm$ 1.3
Method	Average dice of each class (10% Labeled data)															Avg.DSC
	Sp	RK	LK	Ga	Es	Li	St	Ao	IVC	PA	RAG	LAG	Du	Bl	P/U	
SS-Net*	71.2	72.4	73.6	43.5	0.0	82.7	47.9	76.8	70.8	19.3	0.0	0.0	0.0	29.5	1.6	39.29 $\pm$ 1.7
DHC*	87.6	86.3	87.6	48.2	59.1	90.1	65.4	86.5	75.3	66.2	42.6	37.4	44.2	56.9	44.2	65.17 $\pm$ 1.1
ICL	88.3	85.2	86.1	49.2	57.2	89.4	62.7	88.6	76.7	62.7	36.7	31.9	42.1	45.8	41.2	62.92 $\pm$ 0.9
ACTION++	86.5	87.2	88.2	48.7	61.5	90.6	64.2	87.3	79.2	67.5	36.5	27.1	29.5	46.5	38.5	62.60 $\pm$ 1.2
MidSS	88.7	88.1	86.8	49.2	61.2	91.2	66.1	87.1	78.2	64.1	31.2	32.5	23.9	43.7	29.6	61.44 $\pm$ 1.4
MMSeg(ours)	89.2	87.8	89.3	51.3	63.7	89.4	65.3	89.1	78.7	69.4	46.8	40.2	45.3	57.8	46.8	67.34 $\pm$ 1.9

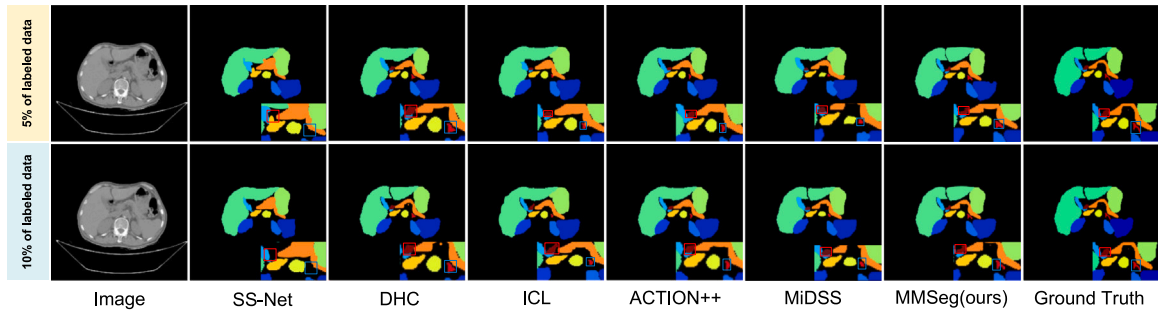


Fig. 4. Visual comparisons of segmentation results on the AMOS Datasets. The bottom-right corner of each segmentation result provides a magnified view of the segmentation details, where the red and blue boxes highlight the boundary details of the duodenum (Du) and left adrenal gland (LAG), respectively.

categories like the Bl and P/U. The experimental results demonstrate that the MMSeg method can achieve efficient segmentation effects with a very low proportion of data labeling and that its performance can continue to improve as more labeled data becomes available. This proves that MMSeg is not only suitable for resource-limited medical image segmentation tasks, but its multitask self-supervised learning mechanism is particularly effective in handling class imbalance and complex medical image structures. Additionally, as shown in Fig. 4, we visualized the segmentation results of MMSeg, showcasing its performance in minority class segmentation.

As shown in Fig. 4, the visualization results of the MMSeg method on the AMOS dataset demonstrate that our approach performs significantly better on the minority classes, Bl and P/U, highlighted within the red box. Compared to other state-of-the-art models, MMSeg effectively leverages unlabeled data to balance the ratio between minority and majority classes, while using multi-task learning to enhance the model's ability to learn representations for minority classes in the images.

Compared to the Synapse dataset, the AMOS dataset faces more severe challenges with minority classes in medical image segmentation. Due to the unique multiple minority class issues and the scarcity of labeled data for these categories in the AMOS dataset, the difficulty of segmentation tasks significantly increases. In this context, the MMSeg method has demonstrated its outstanding performance, especially prominent in handling minority class segmentation tasks. MMSeg not only shows efficiency in utilizing increased annotated data, which is crucial for dealing with class imbalances in large-scale medical image datasets, but its multi-task self-supervised learning mechanism also significantly enhances the recognition of minority classes, improving the model's generalization performance. Additionally, MMSeg has

expanded the minority class labeled data through a teacher–student model, further enhancing the overall segmentation accuracy and model robustness. Thus, MMSeg achieves efficient and precise segmentation results on the extremely imbalanced AMOS and Synapse datasets, demonstrating its significant performance in the field of medical image segmentation.

4.4. Ablation studies

To verify the effectiveness of the key components in our method, we conducted a series of ablation experiments on the key components proposed using the Synapse Datasets, which contain only 20% labeled data. We designed four different ablation experiment settings by incrementally introducing key components.

(1) Baseline: Our baseline method uses the U-net framework.

(2) Setting1: We added a multi-task self-supervised model to the Baseline method.

(3) Setting2: We added the teacher–student model to the Baseline method.

(4) MMSeg: Our complete approach integrates the multi-task self-supervised learning method we proposed into the Setting3 method.

4.4.1. Analysis of critical components through ablation

As shown in Table 3, we conducted a paired t-test statistical significance analysis for the performance differences between each variant and the baseline method. All p-values in the table are less than 0.05, indicating that the variants incorporating key components significantly outperform the baseline method in segmentation performance. It is noted that the baseline segmentation algorithm, without any additional

Table 3

Conducting ablation studies on critical components for medical image segmentation tasks using Synapse data, with 20% of the data labeled. Results are reported in the ‘mean $\pm$ std’ format over 3 repeated experiments. The best results are boldfaced.

Settings	Critical components			Average dice of each class													Avg.DSC	p-value
	L_{mts}	L_{seg}	L_{KL}	Sp	RK	LK	Ga	Es	Li	St	Ao	IVC	PSV	Pa	RAG	LAG		
(1)Baseline	×	✓	×	50.8	67.9	54.2	59.3	3.6	76.5	7.2	63.3	52.4	0.0	6.2	3.5	0.0	34.22 ± 2.7	–
(2)Setting1	✓	✓	×	54.3	69.1	56.6	62.7	11.4	86.2	19.4	68.2	62.2	15.2	17.9	18.8	8.7	42.36 ± 1.8	2.6e−4
(3)Setting2	×	✓	✓	51.5	68.2	56.9	64.0	8.7	85.1	21.7	65.7	59.3	13.1	14.8	23.6	6.5	41.47 ± 2.3	3.2e−4
(4)MMSeg	✓	✓	✓	63.7	71.3	61.3	66.8	18.2	88.7	36.1	70.8	65.7	36.6	32.1	28.3	12.2	50.14 ± 1.8	1.2e−4

Table 4

Exploring the impact of various tasks in a multi-task self-supervised module on medical image segmentation performance in a 20% synapse labeled data environment. Results are reported in the ‘mean $\pm$ std’ format over 3 repeated experiments. The best results are boldfaced.

Method	Multi-task self-supervised				Average accuracy	
	L_{rot}	L_{inp}	L_{dns}	L_{ctr}	DSC $\uparrow$	HD $\downarrow$
Baseline	×	×	×	×	34.22 $\pm$ 2.7	40.82 $\pm$ 3.5
L_{rot}	✓	×	×	×	38.64 $\pm$ 0.8	37.88 $\pm$ 2.9
L_{inp}	×	✓	×	×	37.87 $\pm$ 1.3	32.76 $\pm$ 1.9
L_{dns}	×	×	✓	×	38.36 $\pm$ 1.3	37.57 $\pm$ 2.1
L_{ctr}	×	×	×	✓	39.11 $\pm$ 0.5	34.33 $\pm$ 2.8
$L_{rot} + L_{inp}$	✓	✓	×	×	41.92 $\pm$ 2.8	29.49 $\pm$ 3.6
$L_{rot} + L_{dns}$	✓	×	✓	×	42.96 $\pm$ 3.2	32.66 $\pm$ 1.3
$L_{rot} + L_{ctr}$	✓	×	×	✓	44.82 $\pm$ 1.7	27.39 $\pm$ 2.8
$L_{inp} + L_{dns}$	×	✓	✓	×	40.75 $\pm$ 2.6	28.77 $\pm$ 4.9
$L_{inp} + L_{ctr}$	×	✓	×	✓	44.48 $\pm$ 1.1	27.38 $\pm$ 3.2
$L_{dns} + L_{ctr}$	×	×	✓	✓	46.02 $\pm$ 1.8	28.72 $\pm$ 4.1
$L_{rot} + L_{inp} + L_{dns}$	✓	✓	✓	×	47.14 $\pm$ 1.4	24.94 $\pm$ 3.5
$L_{rot} + L_{dns} + L_{ctr}$	✓	×	✓	✓	46.27 $\pm$ 1.6	25.01 $\pm$ 2.9
$L_{rot} + L_{inp} + L_{ctr}$	✓	✓	×	✓	47.28 $\pm$ 1.2	25.13 $\pm$ 2.6
$L_{inp} + L_{dns} + L_{ctr}$	×	✓	✓	✓	45.29 $\pm$ 0.6	24.80 $\pm$ 3.2
MMSeg(ours)	✓	✓	✓	✓	50.14 $\pm$ 1.8	21.03 $\pm$ 3.1

modules, did not perform well on minority classes such as the Es, St, Pa, RAG, and LAG. The Setting1 method, by introducing a multi-task self-supervised module into the baseline, successfully increased the average DSC by 23%. This significant improvement is attributed to our multi-task self-supervised learning strategy, which enables the segmentation model to deeply mine the representational information from both unlabeled and labeled data. Specifically, Setting1 utilizes a multi-task learning framework that facilitates a comprehensive understanding of different tissue structures, thus achieving significant performance enhancements in the segmentation of minority classes. The Setting2 method adopts a teacher–student module strategy within the semi-supervised learning framework on top of the baseline model. This strategy effectively expands the sample pool for minority classes through pseudo-labels generated by the teacher network, significantly enhancing the model’s ability to learn these classes, particularly in the segmentation tasks of the RAG and Pa. Ultimately, our complete MMSeg method significantly improved segmentation performance across all areas, especially in the segmentation of minority classes, achieving the highest average DSC of 50.14 $\pm$ 1.8. This method significantly enhances the model’s ability to recognize features of minority classes and greatly improves the accuracy and reliability of segmentation.

4.4.2. Effectiveness of multi-task self-supervised module

In Table 4, we detail the effectiveness of incorporating various combinations of loss functions within a multi-task self-supervised module for medical image segmentation. The results illustrate how each

Table 5

Analysis of hyperparameters for Pseudo-label in the MMSeg Semi-Supervised Module. Results are reported in the ‘mean $\pm$ std’ format over 3 repeated experiments. The best results are boldfaced.

Hyperparameter	Average accuracy	
K	DSC	HD
3	32.75 $\pm$ 0.7	42.88 $\pm$ 0.9
5	44.27 $\pm$ 1.2	30.37 $\pm$ 1.7
8	50.14 $\pm$ 1.8	21.03 $\pm$ 3.1
10	50.07 $\pm$ 1.3	22.15 $\pm$ 2.4

combination influences segmentation performance, particularly focusing on two critical metrics: DSC and the Hausdorff Distance (HD). The baseline model, which solely utilizes the standard segmentation loss function, shows an average DSC of 34.22 $\pm$ 2.7 and an HD of 40.82 $\pm$ 3.5. This sets a foundational benchmark for assessing the additional benefits derived from integrating self-supervised tasks. Upon introducing a single self-supervised loss function (L_{rot} , L_{inp} , L_{dns} , L_{ctr}), there is a notable improvement in performance across both metrics, especially in HD. This reduction in maximum error highlights that self-supervised learning substantially enhances the model’s capability to refine image boundaries and intricate details. Combining two or more loss functions further amplifies these benefits, with significant enhancements observed particularly in the DSC values. This suggests that the multi-task learning strategy not only effectively leverages unlabeled data to enrich the model’s learning but also fosters a more holistic enhancement across various segmentation challenges. Remarkably, when all four loss functions L_{rot} , L_{inp} , L_{dns} , and L_{ctr} are simultaneously employed, the model achieves its best performance: the highest average DSC of 50.14 $\pm$ 1.8 and the lowest HD of 21.03 $\pm$ 3.17. This result emphasizes the importance of integrating multiple self-supervised tasks in enhancing the model’s accuracy and reliability in dealing with complex structures in medical images.

4.4.3. Effectiveness of semi-supervised module

We analyzed the impact of pseudo-label generation iterations on the performance of the MMSeg framework. MMSeg employs a semi-supervised learning module based on a teacher–student model structure, generating pseudo-labels through an iterative process. We investigated the effect of the number of pseudo-label iterations on segmentation accuracy and boundary precision, conducting experiments with different values of K for the iterations. The experimental results are shown in the Table 5. When the number of iterations K = 3, the model’s DSC is 32.75, and the HD is 42.88. Due to the fewer iterations, the generated pseudo-labels may lack sufficient accuracy, limiting the effective guidance for the student model. However, as the number of iterations K increases, the model performance improves significantly, with DSC increasing to 44.27 and HD decreasing to 30.37. This result indicates that more iterations enable the teacher model to generate more accurate pseudo-labels, thereby enhancing the learning effect of the student model. When the number of iterations K = 8, the model achieves the best performance, with DSC rising to 50.14 and HD dropping to 21.03. This suggests that at this number of iterations,

Table 6

Analysis of dynamic weight control mechanism and progressive learning strategy. Results are reported in the ‘mean $\pm$ std’ format over 3 repeated experiments. The best results are boldfaced.

Strategy		Average accuracy	
Dynamic weight control	Progressive learning	DSC $\uparrow$	HD $\downarrow$
×	×	38.06 $\pm$ 2.3	37.61 $\pm$ 1.5
✓	×	43.14 $\pm$ 2.7	28.06 $\pm$ 2.3
×	✓	41.62 $\pm$ 1.8	27.52 $\pm$ 2.8
✓	✓	50.14 $\pm$ 1.8	21.03 $\pm$ 3.1

the accuracy of the pseudo-labels and the guidance effect on the student model reach an optimal balance. However, when the number of iterations increases to $K = 10$, although the performance remains high, with DSC at 50.07 and HD at 22.15, it shows a slight decrease compared to $K = 8$. This may indicate that too many iterations could lead to diminishing returns or even cause the student model to overfit the pseudo-labels generated by the teacher model.

4.4.4. Effectiveness of dynamic weight control mechanism and progressive learning strategy

In our research on multimodal medical image segmentation MMSeg framework, we deactivated the teacher–student model within the semi-supervised learning module to stabilize the training process, which allowed us to investigate the effectiveness of the dynamic weight control mechanism and the progressive learning strategy. According to the data shown in Table 6, applying either dynamic weight control alone achieving a DSC of 43.14 ± 2.7 and an HD of 28.06 ± 2.3 or progressive learning strategies alone achieving a DSC of 41.62 ± 1.8 and an HD of 27.52 ± 2.8 significantly enhances the model’s segmentation performance compared to the baseline model without any strategies DSC of 38.06 ± 2.3 and HD of 37.61 ± 1.5 . When these two strategies are combined, the model performs optimally, significantly improving the DSC to 50.14 ± 1.8 and reducing the HD to 21.03 ± 3.1 . These results fully demonstrate that dynamic weight control and progressive learning strategies can work synergistically to optimize the training process of MMSeg. The dynamic weight control mechanism effectively maintains the model’s ability to generalize across different tasks by preventing any single task from dominating the multi-task learning environment, thereby preserving previously learned feature information. Simultaneously, the progressive learning strategy incrementally enhances the model’s focus on segmentation accuracy throughout the training process, which is crucial for handling the varied quality and quantity of labels in real-world data.

4.5. Analysis for hyperparameters

In this section, we conducted a detailed analysis of the sensitivity of various hyperparameters in the MMSeg model, focusing on the dynamic weight control mechanism and progressive learning strategy to assess their impact on model performance.

4.5.1. Dynamic weight control mechanism hyperparameter analysis

We conducted a hyperparameter analysis of the dynamic weight control mechanism, as shown in Fig. 5. The experimental results indicate that our method (red curve) consistently achieved higher DSC values and lower HD values across most epochs. In contrast, the other three curves represent the results under fixed weight hyperparameters. As the number of epochs increases, the DSC values of all curves gradually stabilize, while the HD values continue to decrease. Between 100 and 200 epochs, there is a notable improvement, after which the performance gains slow down, indicating that the model is approaching convergence during this stage.

4.5.2. Progressive learning strategy hyperparameter analysis

We conducted an experimental analysis of the hyperparameter β in the progressive learning strategy. As shown in Fig. 6, in Figure (a), DSC increases progressively under all three settings as the number of training epochs increases. Although the DSC of the progressive learning strategy is slightly lower than that of fixed hyperparameter settings ($\beta = 0.25$ and $\beta = 0.5$) in the early training phase (within 50 epochs), it gradually surpasses the other methods as training progresses. By 400 epochs, the DSC reaches 50, significantly outperforming the strategies with fixed hyperparameters. Figure (b) illustrates the trend of the HD. It can be observed that HD decreases rapidly in the initial stages of training, and the progressive learning strategy demonstrates a faster convergence speed compared to fixed $\beta = 0.25$ and $\beta = 0.5$. By the 200 epoch, the HD of the progressive learning strategy has decreased to nearly 30, while the HD for $\beta = 0.25$ and $\beta = 0.5$ remains at a relatively high level.

5. Conclusion

In this paper, we introduce a novel imbalanced medical image segmentation method based on a multi-task learning framework. This method combines self-supervised learning and semi-supervised learning to enhance the model’s capability to address class imbalance in unlabeled data. In our multi-task framework, the self-supervised task facilitates representation learning from unlabeled data to mitigate feature extraction biases caused by imbalances in regions of interest. The semi-supervised task efficiently leverages a small amount of labeled data along with a large volume of unlabeled data to generate high-confidence pseudo-labels. Furthermore, we have designed a progressive learning strategy and a dynamic weight adjustment mechanism to prevent potential fitting challenges during joint training and to avoid feature representation biases caused by inconsistent training speeds among multiple tasks. Extensive experiments on two imbalanced abdominal multi-organ segmentation tasks demonstrate that our method achieves state-of-the-art performance.

CRediT authorship contribution statement

Feiyang Yang: Methodology, Formal analysis, Data curation, Conceptualization. **Xiongfei Li:** Writing – review & editing, Resources, Conceptualization. **Bo Wang:** Writing – original draft, Validation, Software, Methodology, Formal analysis. **Tianyang Zhang:** Validation, Resources, Data curation. **Xinyi Yu:** Writing – review & editing, Visualization, Validation, Formal analysis. **Xingcheng Yi:** Investigation, Formal analysis, Data curation. **Rui Zhu:** Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

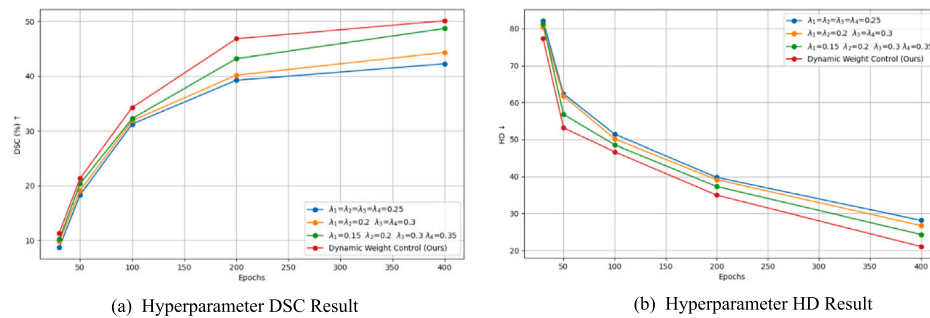


Fig. 5. Result curves of the dynamic weight control mechanism under different weight hyperparameters: (a) DSC result curve, (b) HD result curve.

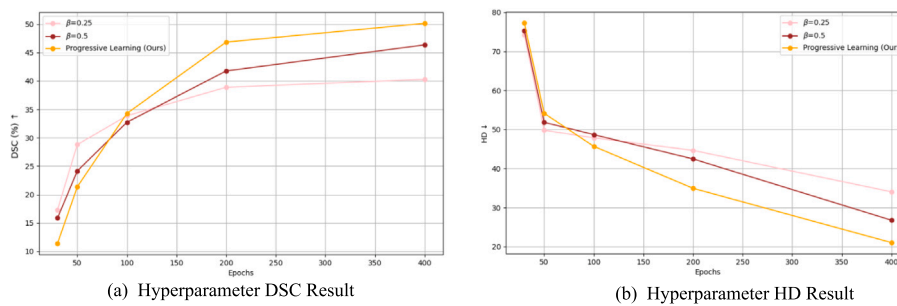


Fig. 6. Result curves of the progressive learning strategy under different weight hyperparameters: (a) DSC result curve, (b) HD result curve.

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